

Week Seven Technical Report

Introduction to React and Its Components

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Teach2Give QA & QE Training Program

Overview

Week 7 introduced us to frontend development with React.js. The main focus was understanding the fundamentals of React, including its component-based architecture, the use of JSX, props, and state management. These concepts form the foundation for building dynamic and reusable user interfaces.

Training Modules Covered

We began by setting up a React development environment using Vite, which provided a fast and efficient build setup. I explored functional and class-based components and learned how to structure them for better scalability.

The lessons covered JSX syntax, which allows embedding HTML-like elements within JavaScript, and how React's virtual DOM efficiently updates the UI. We also discussed how props enable data sharing between components and how state helps manage dynamic data and user interactions.

Practical Exercise and Project

To reinforce these concepts, we built simple React interfaces, applying the principles of components, props, and state. This exercise demonstrated how small, reusable pieces of UI can be composed into complete applications.

In our group project—the Bug Tracking System, we are preparing to integrate a React-based frontend to interact with the backend developed in previous weeks. This marked the start of applying React in a real-world project setting.

Challenges Faced

One of the main challenges this week was adapting to the new way of thinking in React's component based architecture compared to traditional HTML and JavaScript. Understanding the flow of data between parent and child components through props also required practice. However, by following examples and experimenting with small components, I became more comfortable with React's logic and reactivity model.

Recommendations

I recommend that future trainees dedicate time to practising component creation and data flow through props and state. Watching React's rendering process in the browser's developer tools is also helpful for understanding how the virtual DOM works. Consistently building small projects before moving to large ones helps strengthen foundational concepts.

Conclusion

This week provided a solid foundation for frontend development using React.js. I learned how to create, structure, and manage components, use props and state effectively, and understand how React updates the UI. These skills will be essential as we continue building the Bug Tracking System's frontend in the coming weeks.

Signature:

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